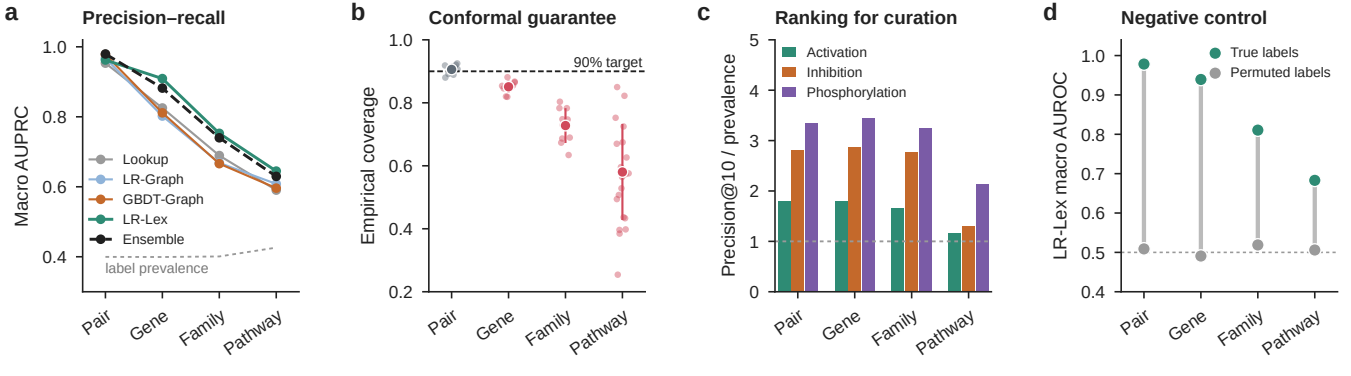
a
Precision–recall
Macro AUPRC
1.0
0.8
0.6
0.4
Lookup
LR-Graph
GBDT-Graph
LR-Lex
Ensemble
label prevalence
Pair
Gene
Family
Pathway
b
Conformal guarantee
Empirical coverage
90% target
1.0
0.8
0.6
0.4
0.2
Pair
Gene
Family
Pathway
c
Ranking for curation
Precision@10 / prevalence
Activation
Inhibition
Phosphorylation
5
4
3
2
1
0
Pair
Gene
Family
Pathway
d
Negative control
LR-Lex macro AUROC
True labels
Permuted labels
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Pair
Gene
Family
Pathway